Reinforcement Learning-Based Sequential Parameter Tuning for Image Signal Processing

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***Abstract*—Hardware image signal processing (ISP) transforms RAW inputs into high-quality RGB images through a series of processing modules, each with numerous tunable parameters. Traditionally, these parameters are manually tuned by imaging experts, a time-consuming and subjective process. Recent deep learning approaches predict ISP parameters, but often treat the process as a black box and overlook the intrinsic relationships among ISP modules. To address these fundamental issues, we introduce a novel ISP parameter optimization model based on single-agent reinforcement learning (RL) (*i.e.*, SARL-ISP), formulating the hardware ISP parameter tuning as a sequential optimization problem. During the optimization process, the agent updates ISP parameter tuning strategies for different tasks through interaction with the environment. In order to explore the influence of the sequential structure of hardware ISP modules and the coupling relationships among ISP parameters on the tuning process, we further propose a sequential ISP framework based on collaborative multi-agent RL (*i.e.*, MARL-ISP). Specifically, the serialized parameter tuning module (SPTM) realistically simulates the process of manual prediction and module pipeline. Additionally, the feature selection module (FSM) facilitates the transmission and fusion of agent features, thereby selecting more appropriate feature inputs for downstream tasks. Extensive experiments across various tasks (*e.g.*, object detection, instance segmentation) validate the effectiveness and efficiency of our models. Even with minimal training data, our models also outperform current state-of-the-art methods in both quantitative metrics and qualitative evaluations.**

***Index Terms*—Low-level vision, reinforcement learning, image signal processing, sequential parameter tuning.**

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I. Introduction

HARDWARE Image Signal Processors (ISPs) are low-level image signal processing pipelines designed to convert RAW images into RGB format for human viewing and machine vision. Hardware ISPs offer advantages such as superior image quality and shorter processing times compared to software-based alternatives. Consequently, they are extensively applied in various domains, including smartphone cameras [1], [2], video surveillance systems [3], robotics [4], and autonomous driving technologies [5].

The ISP pipeline serves as a crucial bridge between image sensors and downstream applications [9], [14]. It consists of a series of interconnected processing modules [6], each dedicated to specific sub-tasks like denoising and white balance. Each module contains a set of adjustable parameters [8] that intimately impact the final image quality. The predominant practice in the industry relies on image experts [7] to manually fine-tune these parameters for various scenarios, as depicted in Fig. 1(a). Nevertheless, this approach is time-consuming, labor-intensive, and occasionally susceptible to subjective bias.

To tackle ISP optimization, three mainstream approaches have emerged: ISP proxy-based methods, ISP parameter prediction-based methods, and ISP parameter tuning-based methods. The ISP proxy-based methods [10], [11] typically train a Convolutional Neural Network (CNN) as an implicit ISP proxy to generate RGB images from RAW images directly, bypassing the ISP parameter tuning process. The ISP parameter prediction-based methods [17], [22] regard the high-dimensional parameter space as a global black box space and train a neural network to predict ISP parameters. ISP parameter tuning-based methods are currently dominated by Covariance Matrix Adaptation Evolutionary Strategies (CMA-ES) [14], [15], a 0^{th}-order stochastic evolutionary search method that optimizes ISP parameters through multi-objective scalarization. Although past years have witnessed great development of the above methods, they still have the following limitations:

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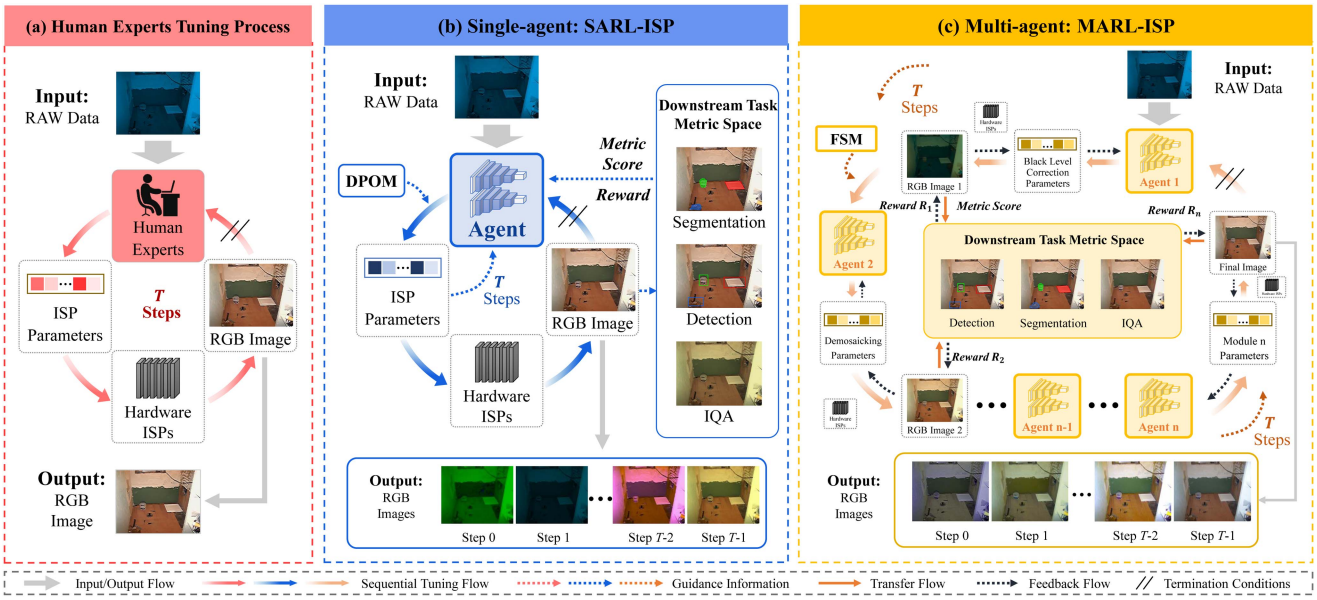

Fig. 1. Comparison between expert tuning and our methods. (a) Human experts tuning process. The image experts iteratively tune the ISP parameters through empirical perception. (b) SARL-ISP predicts ISP parameters under the guidance of feedback from various downstream applications. (c) MARL-ISP optimizes ISP parameters step by step under the feedback of specific downstream tasks and final image quality based on ISP serialization structure.

- ISP proxy-based methods fail to deliver specific ISP parameters for each image, leading to poor performance across different tasks. Moreover, these methods require large amounts of training data, which is often unavailable for limited RAW-RGB datasets.
- ISP parameter prediction-based methods predict all parameters at once, overlooking the correlation among ISP modules. Certain approaches are limited to optimizing one module at a time, rather than jointly optimizing multiple ISP modules.
- The CMA-ES-based ISP parameter tuning method is essentially a space search method without the training stage. It is extremely time-consuming to search for the optimal parameters for each image. In addition, the model tends to get stuck in local optima.

In order to overcome these limitations, we propose SARL-ISP, a novel ISP parameter tuning model based on single-agent Reinforcement Learning (RL), illustrated in Fig. 1(b). Our model leverages the remarkable efficacy of RL in solving sequential decision-making to optimize ISP parameters, demonstrating significant potential in this domain. Specifically, the model trains a single-agent network to optimize ISP parameters over multiple steps. Similar to human experts in parameter tuning, the agent adjusts parameters based on environmental feedback from the previous step until the optimization converges. To guide the agent to make decisions, we develop a Dual-branch Parameter-guided Feature Fusion Module (DPFFM) to extract and utilize environmental information. Furthermore, to prevent parameters from converging to local optima during ISP parameter tuning [41], we propose a Dynamic Parameter Optimization Module (DPOM), which dynamically updates ISP parameters by weighing and aggregating the current parameters, agent decisions, and feedback.

While SARL-ISP incrementally optimizes the ISP parameters, the agent adjusts the parameters for all modules simultaneously at each step. This is due to the limitation of the single agent structure, neglecting the sequential information of ISP modules and the correlations among parameters within the same module. However, these factors are crucial in practical image processing and parameter tuning. Upon examining ISP's sequential structure, we realize that a multi-agent RL architecture can effectively incorporate the sequential information that SARL-ISP overlooks. Therefore, we further present MARL-ISP, a sequential ISP parameter tuning framework based on collaborative multi-agent RL, as illustrated in Fig. 1(c). The MARL-ISP accommodates larger observation and action spaces, making it more suitable for large numbers of parameters in the ISP. In addition, we propose two innovative modules in the MARL-ISP: the Serialized Parameter Tuning Module (SPTM) restores the process of module pipeline and the Feature Selection Module (FSM) provides better feature representations for the model. During training, each agent can learn both its own strategy and the behaviors of other agents, facilitating collaborative learning. Furthermore, the MARL-ISP has the flexibility of adding or removing agents to accommodate various hardware ISPs.

The contributions of this paper are summarized as follows:

- *Pioneering a single-agent ISP parameter tuning model:* We introduce SARL-ISP, a novel single-agent reinforcement learning model for optimizing ISP parameters. SARL-ISP leverages current parameters and feedback performance from downstream tasks to guide the search for globally optimal parameter results.
- *Proposing a multi-agent sequential ISP framework:* We further propose MARL-ISP, a collaborative multi-agent framework for sequential ISP parameter tuning. The SPTM explores the impact of sequence information of ISP

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modules and the intrinsic relationship of parameters during tuning. The FSM provides agents with fused features that are suitable for different downstream tasks.

- *Demonstrating superior performance:* A large number of experimental results validate the effectiveness of our models, which outperform other state-of-the-art approaches in various tasks, including object detection and instance segmentation. In particular, our models achieve competitive performance even with limited training data.

II. Related Work

A. ISP Proxy-Based Methods

Recently, a series of ISP proxy-based methods have been proposed to tackle the ISP optimization problem. Some approaches try to model the traditional hardware ISP pipeline using an approximate CNN proxy model, proposing an end-to-end deep neural model to optimize the proxy network [1], [20], [21]. For example, Ignatov et al. [52] replace a mobile camera ISP with a deep learning model to solve complex hand-crafted camera ISP situations. Other works replace specific ISP pipeline modules with learnable differential models [11], [13]. For instance, Chen et al. [11] propose a differentiable approximation model to perform low-light image processing in the denoising task. Tseng et al. [13] optimize the entire optical system, ISP, and downstream network together using a task-specific loss function. Cui et al. [24] propose an unsupervised RAW-to-RAW mapping model to achieve cross-camera color space alignment. Zhang et al. [27] present ISPDiffuser, a diffusion-based ISP proxy that decouples RAW-to-RGB mapping into grayscale detail reconstruction and color consistency mapping. These methods directly map RAW to RGB images, removing the configuration of ISP parameters. However, they adopt fixed network parameters for all inputs, limiting their adaptability to diverse scenes.

B. ISP Parameter Optimization-Based Methods

ISP parameter optimization-based methods can be divided into two categories: ISP parameter prediction-based methods [12], [13] and ISP parameter tuning-based methods [14], [15]. The former aims to optimize image quality by directly predicting parameters relevant to the ISP. For instance, Tseng et al. [12] devise a CNN proxy for the black-box image processor to mimic the hardware ISP. Qin et al. [17] use an attention-based CNN method to infer all ISP parameters. Cui et al. [25] leverage two types of adapters to more effectively integrate camera RAW images with sRGB pre-trained models, thereby reducing the gap between the two domains. However, these methods require a large amount of training data, which is scarce in available RAW images. Nishimura et al. [16] optimize ISP components with sequential prior knowledge based on a 0-th order Nelder-Mead method. However, it only applies to a single ISP module, limiting its applicability to hardware ISPs. Qin et al. [18] utilize the similarity among parameters to group them and guide the model sequence process. But it needs to be repeatedly migrated and retrained across different ISP lens sensors, which is very costly.

The ISP parameter tuning-based methods exploit different parameter space search algorithms to optimize hardware ISP parameters. For example, Mosleh et al. [14] propose a novel CMA-ES strategy to optimize hardware ISP parameters by multi-objective scalarization and initial search space reduction. Robidoux et al. [15] formulate the HDR ISP parameter search as an end-to-end optimization problem and propose a mixed 0th and 1st-order block coordinate descent optimizer to learn the network weights across multiple modules jointly. Although practical for small datasets, these methods require re-search for each new image, which is extremely time-consuming.

C. Reinforcement Learning (RL)

1) Preliminaries of RL: The core of RL involves two major interacting entities: the agent and the environment. The agent observes the state $s_t \in S$ at each time step t and takes an action a_t. The action influences the environment, leading to an updated state s_{t+1} and a reward signal $r_t \in R$ that is returned to the agent. The agent's ultimate goal is to learn an optimal policy π^* that guides action selection to maximize the accumulated reward $\mathbf{R}_t = \sum_{j=0}^{T} \gamma^j r_{t+j}$, where r_{t+j} is the reward at the time step $t+j$ and $\gamma \in (0, 1]$ is the discount factor that determines the weight given to future rewards.

In RL, the state-action value function, $Q^{\pi}(s_t, a_t)$, represents the expected cumulative reward from state s_t and action a_t under policy π. It is widely used for action evaluation and policy improvement. The state value function $V^{\pi}(s_t)$ is a function that evaluates the expected cumulative reward starting from state s_t under policy π. Its purpose is to assist in the evaluation and improvement of the policy.

The Actor-Critic (AC) algorithm [31] widely used in RL employs two networks: an actor network $\pi_\theta(a_t|s_t)$ parameterized by θ and a critic network parameterized by ϕ to learn the value function $V_\phi^{\pi_\theta}(s_t)$. The policy network is updated by maximizing the following objective function:

$$J^{\pi_\theta} = \mathbb{E}_{\pi_\theta}\left[\log \pi_\theta(a_t|s_t)\mathrm{A}_t\right], \tag{1}$$

where A_t is the advantage function that evaluates a specific action compared to the average level. The value function parameter ϕ is updated by minimizing the following function:

$$J^{V_\phi} = \mathbb{E}_{\mathcal{D}}\left[\left(\mathbf{R}_t + \gamma V_{\phi'}(s_{t+1}) - V_\phi(s_t)\right)^2\right], \tag{2}$$

where $\mathcal{D}$ denotes an experience replay buffer and ϕ denotes the parameters obtained from earlier iterations.

2) Single-Agent of RL: RL is a branch of machine learning used to solve sequential decision-making problems. Classic single-agent algorithms include Q-learning [34], Deep Deterministic Policy Gradient [35], and Proximal Policy Optimization (PPO) [36] algorithms.

Recently, many researchers have investigated RL in low-level image quality enhancement. Bajaj et al. [39] focus on tackling sRGB to sRGB image denoising tasks and propose RSE-RL to enhance decoder transformation to improve denoising performance. Wang et al. [40] use RL to adaptively choose exposure bracketing to obtain a high dynamic range image. Similarly, Shin et al. [41] replace the actual ISP with an ISP toolbox containing

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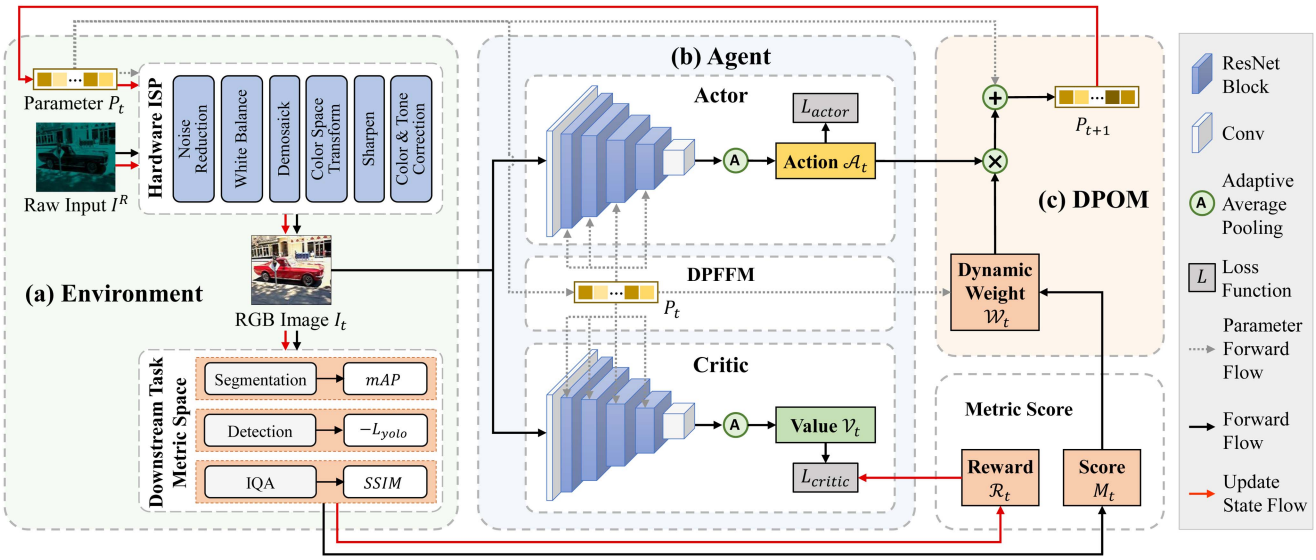

Fig. 2. The overall pipeline of the proposed SARL-ISP model, which tackles the hardware ISP parameter tuning as a progressive parameter optimization problem. (a) At each time step t, taking I^R and P_t as input, the environment generates an RGB image I_t and a metric score M_t. (b) The agent observes the state $\mathcal{S}_t$ (including P_t and I_t) to decide an action $\mathcal{A}_t$ for parameter optimization. (c) The DPOM updates P_t to P_{t+1} according to the trade-off among $\mathcal{A}_t$, P_t, and M_t. The above processes iterate T steps to continuously optimize hardware ISP parameters.

lightweight CNN tools and traditional tools. The agent is trained to select these tools to process RAW images effectively. Wang et al. [26] present a task-driven, scene-adaptive ISP approach that leverages RL to design the optimal ISP pipeline and tune its parameters to maximize detection performance.

3) Multi-Agent of RL: Multi-agent demonstrates greater capability when facing tasks requiring cooperation or competition. Their application domains are broader [29], [30], [42], which involve intricate interactions among multiple parties. Interactive evolution in these systems often leads to the emergence of new behaviors, increasing problem complexity. Several classic multi-agent algorithms [37], [38] are developed to address these challenges. With the widespread application of multi-agent models, the research on image enhancement using multi-agent systems gains more attention. Richard et al. [43] propose a multi-agent framework for data discovery and image enhancement. Rodin et al. [44] present a parallel image processing system based on the concept of reactive agents.

III. Method

In this section, we first present how the hardware ISP parameter optimization problem can be formulated as a continuous Partially Observable Markov Decision Process (POMDP) and provide a detailed problem formulation. For completeness of the article structure and logical coherence, we introduce the single-agent parameter optimization approach (*i.e.*, SARL-ISP) and analyze its limitations. Building on this, we introduce a multi-agent parameter optimization method (*i.e.*, MARL-ISP). We thoroughly discuss the design details of the reinforcement learning tuple for MARL-ISP, along with its innovative modules and training process. Additionally, for the SARL-ISP, we supplement more experimental extensions and verifications outside the conference paper in Section IV.

A. Problem Formulation

The hardware ISP $F_{ISP(;)}$ takes the RAW image I^R from the image sensor and the specific ISP parameters P_t as input, and outputs an RGB image I.

SARL-ISP: The agent optimizes its strategy through continuous interaction with the environment $\mathcal{E}$. Upon receiving actions from the agent, the $\mathcal{E}$ generates an RGB image based on new ISP parameters P. A task-specific metric function $M(\cdot)$ with the environment scores the generated images and calculates the reward $\mathcal{R}_t$. The multi-step optimization process is represented as the tuple $\{\mathcal{S}_t, \mathcal{A}_t, \mathcal{R}_t, \mathcal{S}_{t+1}\}_{t=0}^{T-1}$, where T is the maximum optimization step, $\mathcal{S}_t = \{P_t, I_t = F_{ISP}(I^R; P_t)\}$ is the state at time t, $\mathcal{A}_t$ are the actions based on the current strategy, $\mathcal{R}_t$ is the reward, and $\mathcal{S}_{t+1}$ is the next state.

MARL-ISP: The interaction process between multiple agents and the environment is similar to that of a single agent. The entire sequential optimization process can also be expressed as a new tuple $(\{A_t, S_t, O_t, R_t, S_{t+1}\}_{t=0})$. $O_t^i \in S_t^{global}$ and $A^i \in A_{total}$ represent the local observation value and the actions for each agent i, respectively, while S_t^{global} represents the global state. Each agent follows an Actor-Critic structure. The actor of each agent observes only a portion of the environment E, *i.e.*, the image generated based on the module for which it predicts parameters and the parameters of preceding modules. The critic receives the global observation, that is the final image generated by the entire ISP pipeline.

B. ISP Parameter Tuning Based on Single-Agent Reinforcement Learning (SARL-ISP)

1) Actor-Critic Network: The agent is designed to optimize ISP parameters based on the current state $\mathcal{S}_t$. The agent network consists of an actor network and a critic network. Each network

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follows a similar structure, comprising several sequential modules: a convolutional (conv) layer, multiple residual blocks [65], another conv layer, and adaptive average pooling. The actor network predicts the ISP parameter variation as an action $\mathcal{A}_t \in \mathbb{R}^k$ based on the current state $\mathcal{S}_t$, expressed as:

$$\mathcal{A}_t = \pi(\mathcal{S}_t; \theta_\pi), \tag{3}$$

where $\pi(;\theta_\pi)$ denotes the actor network with weights θ_π. The agent optimizes the ISP parameters P_t using $\mathcal{A}_t$:

$$P_{t+1} = \text{clip}(Norm(P_t) + Norm(\mathcal{A}_t), -1, 1), \tag{4}$$

where clip$(\cdot)$ is a clipping function [36] that normalizes vector values to $[-1, 1]$, $Norm(\cdot)$ denotes a normalization operation. The critic network evaluates the value $\mathcal{V}_t$ of the current state $\mathcal{S}_t$, and the formula is as follows:

$$\mathcal{V}_t = V(\mathcal{S}_t; \theta_V), \tag{5}$$

where $V(;\theta_V)$ represents the critic network with weights θ_V.

2) Reward Function: Our objective is to generate improved ISP parameters P_{t+1} and RGB image I_{t+1}, resulting in enhanced performance $M(I_{t+1})$. To quantify this improvement, we define the reward function $\mathcal{R}(\cdot)$ as the difference in $M(\cdot)$ before and after executing $\mathcal{A}_t$, expressed as:

$$\mathcal{R}_t = M(I_{t+1}) - M(I_t). \tag{6}$$

The accumulated reward $\mathbf{R}_t$ can be calculated based on a round of rewards as:

$$\begin{aligned}\mathbf{R}_t = \mathcal{R}_t + \gamma\mathcal{R}_{t+1} + \cdots \\ + \gamma^{T-t+1}\mathcal{R}_{T-1} + \gamma^{T-t}V(S_T; \theta_V),\end{aligned} \tag{7}$$

where γ denotes the discount factor.

3) Dual-Branch Parameter-Guided Feature Fusion Module (DPFFM): To optimize decision-making, the DPFFM combines the image features F_{I_t} and parameters P_t to generate a fusion feature $\hat{F}_{I_t}$, expressed as:

$$\hat{F}_{I_t} = F_{I_t} \copyright\, \xi(P_t), \tag{8}$$

where © denotes the channel-wise concatenation and $\xi(\cdot)$ is a reshaping operation [12] that replicates P_t to match the same spatial dimension as F_{I_t}. To prevent discrepancies in feature representations at different levels, we also adopt multi-layer feature fusion before each residual block.

4) Dynamic Parameter Optimization Module (DPOM): During optimization, some parameters tend to approach boundary values (*i.e.*, -1 or 1), which can lead the optimized ISP parameters to local optima, a common challenge also observed in RL domains [32], [33]. To address this, we introduce DPOM, which guides parameter optimization by dynamically weighing current parameters and feedback performance from downstream tasks. According to different tasks, the environment provides the corresponding metric functions $M(\cdot)$. For dynamic parameter optimization, we modify (4) as follows:

$$P_{t+1} = \text{clip}(Norm(P_t) + \mathcal{W}_t \times Norm(\mathcal{A}_t), -1, 1), \tag{9}$$

where $\mathcal{W}_t \in \mathbb{R}^k$ denotes a set of adaptive weights, determined by the trade-off between the P_t and the metric score $M(I_t)$, formulated as:

$$\mathcal{W}_t = \alpha(e^{1-|P_t|} - 1) \cdot log\left[\beta\left(M_{tgt}(I_t) - M(I_t)\right) + 1\right], \tag{10}$$

where α and β are two hyperparameters to balance the weight, $1 - |P_t|$ represents the distance to the parameter boundary, $M_{tgt}(I_t)$ denotes the target score.

5) the Training for the SARL-ISP: We train our SARL-ISP model based on the training process of the PPO algorithm [36]. For more details, please refer to the *RL-seqISP* [23].

To optimize the critic network, we aim to minimize the expectation of the difference between its predicted value $\mathcal{V}_t$ and the accumulated reward $\mathbf{R}_t$. The loss function L_{critic} of the critic network is as follows:

$$L_{critic} = \hat{\mathbb{E}}_t\left[(\mathbf{R}_t - V(\mathcal{S}_t; \theta_V))^2\right]. \tag{11}$$

For the actor network, our objective is to maximize accumulated return. The loss function L_{actor} is formulated as follows:

$$\begin{aligned}L_{actor} = \hat{\mathbb{E}}_t\Bigg[\min\Bigg(r_t(\theta_\pi)\hat{\mathrm{A}}_t, \\ \text{clip}\Big(r_t(\theta_\pi), 1-\epsilon, 1+\epsilon\Big)\hat{\mathrm{A}}_t\Bigg)\Bigg],\end{aligned} \tag{12}$$

where ϵ is a hyperparameter of clip$(\cdot)$, $\hat{\mathrm{A}}_t$ indicates an advantage function [66], and $r_t(\theta_\pi)$ is an objective function to constrain the size of actor updates. For detailed descriptions, please refer to [36].

C. Sequential ISP Parameter Tuning Based on Collaborative Multi-Agent Reinforcement Learning (MARL-ISP)

In order to further integrate the sequential information of ISP during the parameter optimization, we introduce a multi-agent approach to address the order problem of ISP parameter tuning in this section. The collaborative multi-agent architecture ensures that each agent corresponds to a specific functional module within the ISP, enabling accurate and sequential prediction of parameters.

1) the RL Tuples' Design of MARL-ISP: The entire sequential optimization process is modeled as a POMDP, which can be described by the tuple $(\{A_t, S_t, O_t, R_t, S_{t+1}\}_{t=0})$. This section details the modeling of actions, states, observations, the design of rewards, and the implementation of innovative modules.

- *Action Space:* A key advantage of the multi-agent approach over the single-agent method is that it reduces the dimensions of action space. The action A_i of agent *i* is the amount of change in the parameters that need to be predicted for the corresponding module. The total action dimension of the system is the sum of each module's action dimensions. The joint action space can be expressed as $A_{total} = \{A^1, A^2, \ldots, A^n\}$, where n represents the number of functional modules in the ISP. It is consistent with the number of agents. Based on the current state S_t and observation O_t^i, each agent *i* determines an action $A_t^i(S_t, O_t^i) \in A_t$ according to the strategy. The final output is the global joint action $A_t = \{A_t^1 \cup A_t^2, \ldots, \cup A_t^n\}$. The new ISP parameter values for time step $t+1$ are obtained

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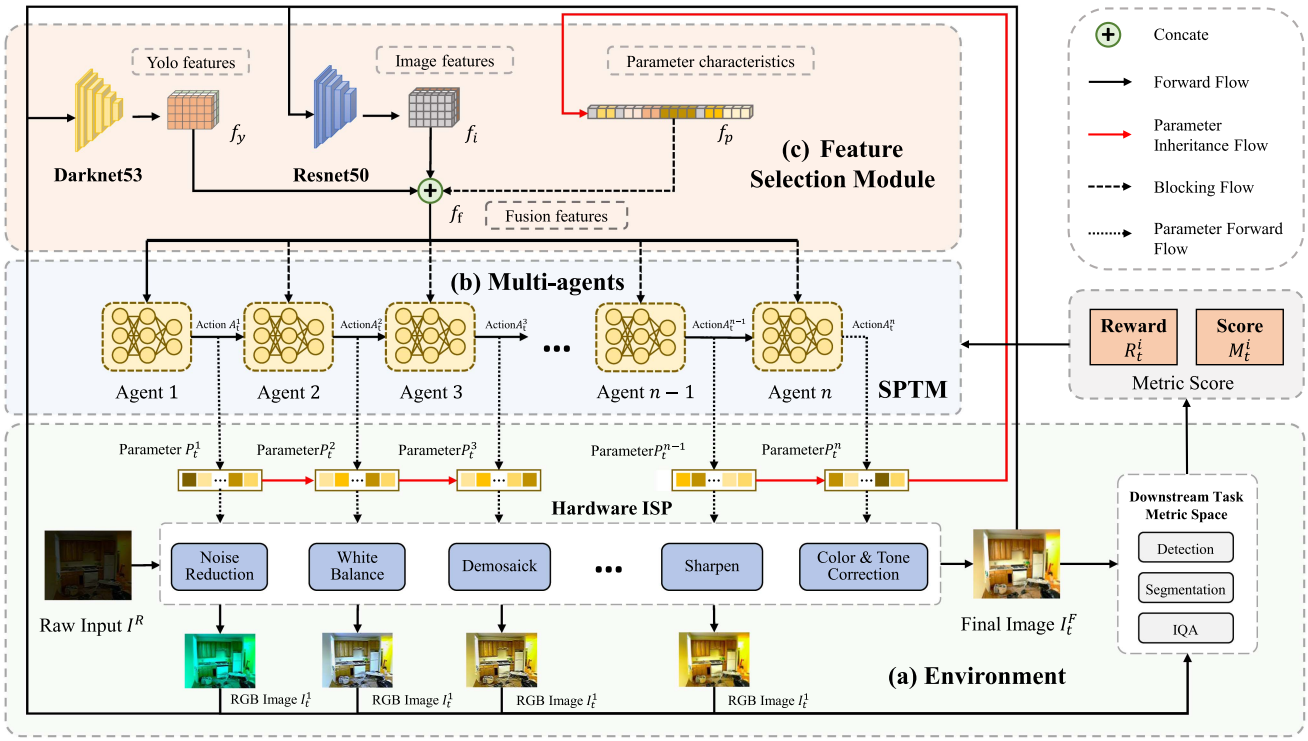

Fig. 3. The overall pipeline of the proposed MARL-ISP model, which is based on the structural serialization of hardware ISP, predicts and optimizes the ISP parameters within each module. (a) At each time step t, the environment takes the I^R and P_t^i as input. Subsequent modules in the ISP will inherit the parameters predicted by the previous modules to generate RGB images I_t^i. The environment will give the RGB images I_t^i, score M_t^i, and reward R_t^i. (b) In the SPTM, each agent determines the actions A_t^i based on the current state $\mathcal{S}_t$ and local observations O_t^i. (c) The FSM selects the most suitable feature combination based on downstream tasks. The above processes iterate T steps to continuously optimize ISP parameters.

by adding the parameter changes of the current round to the previous round's parameters.

- *State Space:* Each agent's strategy is related to the context information. To ensure the effect of training, we use different feature extraction networks as the backbone for different downstream tasks. The global state at time step t, denoted as S_t^{global}, is the feature obtained by passing the image generated after all agents have predicted through the feature extraction network. It should be emphasized that the S_t^{global} is visible to all agents.
- *Observation:* Each decision-making process in the multi-agent is based on the semantic information of the image generated by the previous modules, rather than directly inputting the same original image. In general, observations are usually part of the global state. So we take the characteristics of the image under local prediction parameters as the local observation values O_t^i of different agents. We sequentially use the parameters predicted by the agent to generate the image for feature extraction. For example, the local observation O_t^1 is the feature of the image generated based on the parameters of the first module predicted by the first agent. For parameters corresponding to agents that have not been predicted, default parameter values are adopted.
- *Reward:* We introduce rewards to guide the agent toward achieving the desired behavior. The initial reward is defined as the difference between the score corresponding to the image in this step and the previous one. This reward is shared across all agents to encourage interaction and improve parameter prediction through unified signals. However, this approach is counterproductive, as it interferes with other agents' decisions and fails to reflect individual contributions. Consequently, it becomes difficult to attribute changes in rewards to specific actions.

Given the sequential structure of multi-agent, we design an independent reward for each agent to address the above issue. The reward assigned to agent i at time step t is defined as the gain of $M(\cdot)$ before and after executing A_t^i, formulated as:

$$r_t^i = M_t^i - M_t^{i-1}. \tag{13}$$

The reward reflects the quality of the agent's previous action. To prevent each agent from only considering its reward and ignoring the cooperation during training, we combine the reward of each agent with the total reward of the entire ISP pipeline. The reward formula after executing the entire ISP pipeline is as follows:

$$r_t = M_t - M_{t-1}. \tag{14}$$

This reward mechanism not only reflects the quality of parameters in each module but also guarantees the final result of the ISP. The formula is as follows:

$$R_t^i = r_t^i + \omega * r_t, \tag{15}$$

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where ω is a hyperparameter that controls the step reward proportion in total reward. After T optimization iterations, each agent *i* has a set of rewards, *i.e.*, $\{R_0^i, \ldots, R_{T-1}^i\}$. The accumulated reward $\mathbf{R}_t^i$ for each sequence can be defined as:

$$\mathbf{R}_t^i = \sum_{j=t-T+1}^{t} \gamma^{j-1} * R_j^i, \tag{16}$$

where T represents the desired step length, the discount factor γ takes a value in (0, 1].

- *Policy learning and value learning:* Since the value range of each parameter is distinct, we choose to normalize each parameter within the original value range. We use the Multi-Layer Perceptron (MLP) and pooling modules to form the actor and critic networks. The actor network learns a parameter tuning policy $\pi(A_t^i|S_t, O_t^i)$. The formula is as follows:

$$A_t^i = \pi\left(S_t, O_t^i; \theta_\pi\right). \tag{17}$$

The critic network aims to estimate an accurate state value $V_\pi(S_t, O_t^i)$ to maximize returns, formulated as follows:

$$V_t^i = \pi\left(S_t, O_t^i; \theta_V\right). \tag{18}$$

2) Serialized Parameter Tuning Module (SPTM): In the entire ISP pipeline, each functional module is executed in a certain order. Therefore, predicting all parameters at once does not align with the sequential nature of ISP module processing. To mirror the approach of human experts and the ISP structure, we introduce the SPTM.

We model an agent as an expert responsible for the parameter prediction of a single module. The module execution order is consistent with that of the hardware ISP. The first agent predicts the parameters P_t^1 of the module after receiving the current state S_t and local observation O_t^1. After feature extraction and other operations on the resulting image, the next agent receives the processed information, including the local observation O_t^2 and the state S_t. This process continues until all agents execute, producing the final RGB image I_t.

3) Feature Selection Module (FSM): Considering the diversity of features, we believe that different fusion features will be useful for agents in predicting the parameters. So we propose the FSM. Initially, the FSM combines single features into diverse fusion features, each serving as input to the network to guide the generation of agent actions. Images generated from different feature-derived parameters are scored by downstream tasks. Finally, the highest-scoring result determines the most suitable feature for this task.

Algorithm 1. Training procedure of the MARL-ISP.

Initialize θ_π, the parameters for policy π; θ_V, the parameters for critic V; By Orthogonal initialize [51];
while $step \leq step_{max}$ **do**
  Randomly input a RAW image I^R;
  Randomly initialize ISP parameters P_0;
  τ = [] empty list;
  for $t \in \{0, 1, ..., T-1\}$ **do**
    Update state: $\mathcal{S}_t = \{P_t, I_t = F_{ISP}(I^R; P_t)\}$;
    Calculate feedback score:
    $M_t(step) = M_{dt}\big(F_{dt}(I_t; \theta)\big)$;
    for $i \in \{0, 1, ..., n\}$ **do**
      Obtain Local Observations:
      $O_t^i = F_{extract}(I_t^i = F_{ISP}(I^R; P_t^i))$;
      Calculate feedback score:
      $M_t^i(module) = M_{dt}\big(F_{dt}(I_t^i; \theta)\big)$;
      Forward Actions of Agent i:
      $A_t^i = \pi(S_t, O_t^i; \theta_\pi)$;
      Forward Value of State:
      $V_t^i = \pi(S_t, O_t^i; \theta_V)$;
      Optimize ISP parameters:
      $P_{t+1}^i = P_t^i + \mathcal{W}_t \times A_t^i$;
      Update Local Observation:
      $O_{t+1}^i = F_{extract}(I_{t+1}^i = F_{ISP}(I^R; P_{t+1}^i))$;
      Calculate Reward of Module :
      $r_t^i(mod) = M_t^{i+1}(mod) - M_t^i(mod)$;
    end
    Re-update state:
    $\mathcal{S}_{t+1} = \{P_{t+1}, I_{t+1} = F_{ISP}(I^R; P_{t+1})\}$;
    Calculate Reward of Step:
    $r_t(step) = M_{t+1}(step) - M_t(step)$;
    Calculate Reward :
    $R_t^i = r_t^i(mod) + \omega * r_t(step)$;
    for $i \in \{0, 1, ..., n\}$ **do**
      $\tau_i += \left[S_t, O_t^i, A_t^i, R_t^i, S_{t+1}, O_{t+1}^i\right]$
    end
  end
  Compute advantage estimate $\mathrm{A}_i^{(k)}$ via GAE;
  Compute accumulated reward $\mathbf{R}_t^i$;
  for $i \in \{1, 2, ..., n\}$ **do**
    $\mathbf{R}_t^i \leftarrow R_t^i + \mathbf{R}_{t+1}^i$;
    Gradient Backpropagation:
    $\Delta\theta_\pi^i \leftarrow \nabla_{\theta_\pi^i} L_{actor}(\theta_\pi^i)$;
    Gradient Backpropagation:
    $\Delta\theta_V^i \leftarrow \nabla_{\theta_V^i} L_{critic}(\theta_V^i)$;
    Update θ_π^i with $\Delta\theta_\pi^i$ and θ_V^i with $\Delta\theta_V^i$;
  end
end

D. The Training for the MARL-ISP

Our model is primarily trained using the MAPPO [38] algorithm, with the details presented in Algorithm 1.

The actor network is represented as π_θ. It maps agent observations O_t to either a distribution of actions in a discrete action space, or a mean and standard deviation vector of a multivariate Gaussian distribution, and then samples the action A_t from it. The π_θ is trained to maximize the following objective:

$$L(\theta) = \frac{1}{B \cdot n} \sum_{i=1}^{B} \sum_{k=1}^{n} \min \left(r_{\theta,i}^{(k)} \mathrm{A}_i^{(k)},\ \mathrm{clip}\left(r_{\theta,i}^{(k)}, 1-\epsilon, 1+\epsilon\right) \mathrm{A}_i^{(k)} \right)$$

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$$+\sigma\frac{1}{B\cdot n}\sum_{i=1}^{B}\sum_{k=1}^{n}\mathrm{S}\left[\pi_\theta\left(O_i^{(k)}\right)\right],\ \text{where}\ r_{\theta,i}^{(k)}$$
$$=\frac{\pi_\theta\left(A_i^{(k)}\mid O_i^{(k)}\right)}{\pi_{\theta_{old}}\left(A_i^{(k)}\mid O_i^{(k)}\right)}, \tag{19}$$

where the advantage function $\mathrm{A}_i^{(k)}$ is computed using the GAE [66] method, S represents the policy entropy, and σ is a hyperparameter that controls the entropy coefficient. The first half of the formula is to maximize the advantage function. The latter part is to maximize the entropy of the policy to avoid overly concentrated action distributions.

The critic network, V_θ, aims to maximize the estimated return. The loss function is defined as follows:

$$L(\phi)=\frac{1}{B\cdot n}\sum_{i=1}^{B}\sum_{k=1}^{n}\left(\max\left[\left(V_\phi\left(S_i^{(k)}\right)-\hat{\mathbf{R}}_i\right)^2,\right.\right.$$
$$\left(\mathrm{clip}\left(V_\phi\left(S_i^{(k)}\right),\mathrm{V}_{\phi_{\mathrm{old}}}\left(S_i^{(k)}\right)\right.\right.$$
$$\left.\left.\left.-\varepsilon,V_{\phi_{\mathrm{old}}}\left(S_i^{(k)}\right)+\varepsilon\right)-\hat{\mathbf{R}}_i\right)^2\right]\right), \tag{20}$$

where $\hat{\mathbf{R}}_i$ is the accumulated reward, B is the batch size, and n is the number of agents.

IV. Experiments

A. Experimental Settings

Downstream Tasks: To verify our model's broad applicability, we select a diverse set of downstream tasks. The choice of downstream tasks serves to define clear objectives, establish evaluation metrics, and guide the optimization process in our RL framework. For objective tasks, we select the mainstream object detection [47], [50] and instance segmentation [53] tasks in computer vision for training and evaluation. For the subjective evaluation task, we select the Image Quality Assessment (IQA) task [18], [23] for experiments.

Image Signal Processing Pipelines: Each ISP consists of a series of image processing modules. The aforementioned downstream tasks involve four distinct ISPs. ISP-I has good adaptability and is used in three downstream tasks. In the IQA task, we also use ISP-II and ISP-III, which include more functional modules and parameters, to validate the performance of our model. Due to dataset compatibility issues with the ISP, these two ISPs are not applied in detection and segmentation. Some prior methods utilize ISP-IV to process RAW images in the above tasks. Each ISP for different tasks has a set of initial parameters, which we refer to as the default parameters $\mathcal{P}$. Below, we introduce the specific details of these ISPs.

1) ISP-I. For consistency with prior works [17], [18], [23], we use the same synthetic (simulated) ISP to process the simulated RAW as the upstream module for different downstream tasks. The ISP-I contains 20 ISP parameters in total. It includes the following typical stages, including *Noise Reduction* [56], *White Balance* [57], *Demosaicking* [58], *Color Space Transform* [59], and *Color and Tone Correction* [60]. And the default parameters of ISP-I are $\mathcal{P}_1$.
2) ISP-II. The designation of ISP-II is Qualcomm Spectra 580. It is a hardware ISP employed in advanced consumer smartphones. Our models optimize 114 of all ISP parameters for IQA. Its processing stages also include *Noise Reduction*, *White Balance*, *DigitalGain*, *Demosaicking*, *Color Space Transform*, *Sharpening*, *Color and Tone Correction*, and *Compression*. The default parameters of ISP-II are $\mathcal{P}_2$.
3) ISP-III. The ISP-III is also the Qualcomm Spectra 580 ISP for the IQA task and contains 48 parameters in total. Its image processing stages are as follows: *Noise Reduction*, *White Balance*, *Demosaicking*, *Color Space Transform*, and *Color and Tone Correction*. Its default parameters are $\mathcal{P}_3$.
4) ISP-IV. The ISP-IV is a hardware ISP used in other ISP parameter optimization methods, with pertinent details available in [14], [16]. The default parameters within this ISP are denoted as $\mathcal{P}_4$.

Detailed information on ISP modules and parameters is available in the *Supplementary Material*.

Data Settings: To align with the comparison methods [14], [17], [18] in object detection and instance segmentation, we exploit the method [45] to obtain the simulated RAW images based on the MSCOCO datasets [46]. For IQA, we construct a new dataset based on the SONY IMX766 CMOS sensor. It consists of 251 cases indoors and outdoors. More visualization examples of the above dataset can be found in Section V-D and in the *Supplementary Material*.

Training and Evaluation Metrics: In the training and evaluation process, we are concerned with the performance of generated images on downstream tasks. After experimental comparison, we choose $-Loss_{yolo}$ as the object detection metric and the mean Average Precision (mAP) as the instance segmentation metric. For the subjective IQA task, we use the Structural Similarity (SSIM) [63] over the Peak Signal-to-Noise Ratio (PSNR) [64], since SSIM is more in line with human visual perception. In the final result, we employ a comprehensive set of evaluation metrics in different tasks to compare with other methods.

B. Optimization Results on Object Detection

We first evaluate our RL-based models on object detection. To ensure fairness in comparison, we do not select current state-of-the-art object detection algorithms for the downstream task. Instead, we also use the foundational pre-trained YOLOv3 model [47] as the object detection tool consistent with the mainstream methods [14], [17], [18]. In the training stage, we aim to minimize the L_{yolo} (*i.e.*, maximize negative YOLO loss "$-L_{yolo}$"). In the validation stage, we choose the mAP as a metric to compare with other methods.

We compare our approaches with many ISP parameter tuning methods. Note that "Expert-tuned" refers to the process in which

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TABLE I
QUANTITATIVE COMPARISON OF STATE-OF-THE-ART ISP OPTIMIZATION METHODS FOR OBJECT DETECTION ON MSCOCO, INCLUDING DEFAULT AND EXPERT-TUNED CONFIGURATIONS, AS WELL AS MAINSTREAM METHODS. TRAINING DATA SIZE AND ISP FOR EACH METHOD ARE ALSO PROVIDED.

Methods	ISP	Training Images	mAP@0.5	mAP@0.75	mAP@0.5:0.95
Default $\mathcal{P}_4$	ISP-IV [14]	-	0.150	-	-
Blockwise-tuned [16]		-	0.200	-	-
Hardware-tuned [14]		-	0.390	-	-
Default $\mathcal{P}_1$	ISP-I Sec.IV-A	-	0.341	0.224	0.214
Expert-tuned		-	0.568	0.408	0.377
RAW-Mapping [24]		-	0.582	0.389	0.371
ISP-Diffuser [27]		-	0.598	0.423	0.397
Attention-aware [17]		10,000	0.610	0.441	0.410
Sequential-tuned [18]		10,000	0.628	0.452	0.423
RAW-Adapter [25]		10,000	0.629	0.459	0.426
AdaptiveISP [26]		10,000	0.549	0.401	0.377
SARL-ISP(Ours)		1,000	0.640	0.463	0.437
MARL-ISP(Ours)		800	**0.645**	**0.484**	**0.445**

[1] $\mathcal{P}_4$ and $\mathcal{P}_1$ indicate the default parameters based on ISP-IV [14] and ISP-I (Sec.IV-A), respectively.

six image processing experts from mobile phone manufacturers tune ISP parameters for each RAW image and obtain RGB images. Then, they vote to determine the optimal RGB image and its corresponding ISP parameters. The quantitative results are shown in Table I. It is evident that under the same downstream tasks, datasets, and similar synthetic ISP, both the results of $\mathcal{P}_1$ and our methods outperform the corresponding results of $\mathcal{P}_4$, as well as the results of the "Blockwise-tuned" and "Hardware-tuned". Compared to the default ISP parameters $\mathcal{P}_1$ and the expert-tuned method, both SARL-ISP and MARL-ISP significantly improve the quality of generated images after training. In terms of the mAP@0.5 metric, our methods outperform them by approximately 30% and 8%, respectively. Moreover, our approach also achieves notable improvements over mainstream ISP-related methods. MARL-ISP performs sequential predictions by learning the intrinsic correlation between the preceding and subsequent modules and achieves better results on three test metrics. It is worth emphasizing that both SARL-ISP and MARL-ISP achieve these satisfactory results with less than one-tenth of the training data used by other methods. It demonstrates the great potential of RL-based ISP parameter tuning and the effectiveness of our approaches.

The visualization results are shown in Fig. 4. Expert-tuned images are more in line with subjective human perception. The results of existing methods exhibit missed detections and false detections. In contrast, our results are more accurate and closely aligned with the ground truth, demonstrating that optimizing the ISP parameters significantly improves the performance of object detection. Consequently, the outputs of ISP pipelines are no longer intended solely for human perception but are also expected to meet the diverse requirements of emerging tasks such as detection and segmentation in autonomous driving and robotics.

C. Optimization Results on Instance Segmentation

To validate that our models can adjust the most suitable ISP parameters for different tasks, we further conduct experiments on instance segmentation. We use pre-trained Mask R-CNN [53] as the segmentator to make a fair comparison with mainstream methods [14], [17], [18]. During training and evaluation, we obtain RL scores and rewards based on mAP. The results are shown in Table III.

Compared to the default parameters $\mathcal{P}_1$ and expert-tuned, our models demonstrate mAP@0.5 improvements of over 40% and 16%, respectively. Furthermore, both MARL-ISP and SARL-ISP achieve superior performance over all mainstream ISP-related methods across multiple evaluation metrics. Fig. 4 presents the visualization results on the segmentation task, showing a similar trend in the detection task. Compared with the default parameters, expert-tuned results, and mainstream methods, our results yield more precise segmentation masks with

TABLE II
EXPERIMENTAL RESULTS OF DIFFERENT ISPS FOR THE IMAGE QUALITY ASSESSMENT TASK ON THE IMX766 CMOS SENSOR DATASET

Methods	ISP	SSIM	PSNR
Expert-tuned (GT)	ISP-I Sec.IV-A	1.000	-
Default $\mathcal{P}_1$		0.747	15.432
SARL-ISP(Ours)		0.818	18.950
MARL-ISP(Ours)		**0.889**	**24.659**
Expert-tuned (GT)	ISP-II Sec.IV-A	1.000	-
Default $\mathcal{P}_2$		0.797	31.382
SARL-ISP(Ours)		0.821	32.384
MARL-ISP(Ours)		**0.876**	**35.013**
Expert-tuned (GT)	ISP-III Sec.IV-A	1.000	-
Default $\mathcal{P}_3$		0.901	31.904
SARL-ISP(Ours)		0.952	35.228
MARL-ISP(Ours)		**0.960**	**36.317**

[1] $\mathcal{P}_1$,$\mathcal{P}_2$ and $\mathcal{P}_3$ indicate the default parameters based on ISP-I, ISP-II and ISP-III , respectively.

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Fig. 4. Visualization examples of the results on two visual analysis downstream tasks, *i.e.*, (a) object detection and (b) instance segmentation. The first row shows the ground truth annotations on the MSCOCO validation set. The remaining rows present the results of RGB images processed by different ISP settings, including the default parameters, expert-tuned parameters, mainstream ISP methods, and our method.

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TABLE III
QUANTITATIVE COMPARISON OF STATE-OF-THE-ART ISP OPTIMIZATION METHODS FOR INSTANCE SEGMENTATION ON MSCOCO, INCLUDING DEFAULT AND EXPERT-TUNED CONFIGURATIONS, AS WELL AS MAINSTREAM METHODS. TRAINING DATA SIZE AND ISP FOR EACH METHOD ARE ALSO PROVIDED.

Methods	ISP	Training Images	mAP@0.5	mAP@0.75	mAP@0.5:0.95
Default $\mathcal{P}_4$	ISP-IV [14]	-	0.120	-	-
Hardware-tuned [14]		-	0.320	-	-
Default $\mathcal{P}_1$	ISP-I Sec.IV-A	-	0.227	0.132	0.120
Expert-tuned		-	0.469	0.282	0.271
RAW-Mapping [24]		-	0.517	0.319	0.306
ISP-Diffuser [27]		-	0.553	0.381	0.349
Attention-aware [17]		10,000	0.523	0.334	0.315
Sequential-tuned [18]		10,000	0.542	0.351	0.326
RAW-Adapter [25]		10,000	0.564	0.362	0.337
AdaptiveISP [26]		10,000	0.470	0.301	0.288
SARL-ISP(Ours)		1,000	0.630	0.405	0.361
MARL-ISP(Ours)		800	**0.646**	**0.423**	**0.379**

[1] $\mathcal{P}_4$ and $\mathcal{P}_1$ indicate the default parameters based on ISP-IV [14] and ISP-I (Sec.IV-A), respectively.

fewer boundary errors and artifacts. These visual comparisons further highlight the generalization ability of our method in different downstream tasks.

D. Optimization Results on Image Quality Assessment

The parameter optimization of IQA is different from the objective machine vision tasks. It aims to achieve different imaging preferences aligned with human subjective tasks. Images are constantly optimized toward a rendering, a specific imaging style that primarily serves human visual preferences.

Due to the lack of suitable open-source datasets for IQA, we use a SONY IMX766 CMOS sensor to collect 251 RAW images, including 61 images captured in the laboratory and 190 images captured outdoors. Then, several imaging experts tune the ISP parameters for each RAW image and use the Qualcomm Spectra 580 ISP to generate the reference RGB image, *i.e.*, ground truth (GT), named "Expert-tuned (GT)". For training and evaluation, we divide the dataset into 175 and 76 images (about 7:3). In this task, we directly use SSIM and PSNR to evaluate the consistency between the image generated by our model and the GT image without relying on any network. To demonstrate the generality and robustness of our methods, we conduct experiments on the above dataset on the three ISPs.

The results are shown in Table II. In the three ISPs' experimental results, the MARL-ISP is always better than the default parameters and the SARL-ISP. Fig. 5 shows visual image comparisons. The MARL-ISP produces images with better visual quality, consistent with human preferences in terms of texture details, noise control, and color representation. More experiments and visualizations of the IQA task are provided in the *Supplementary Material*.

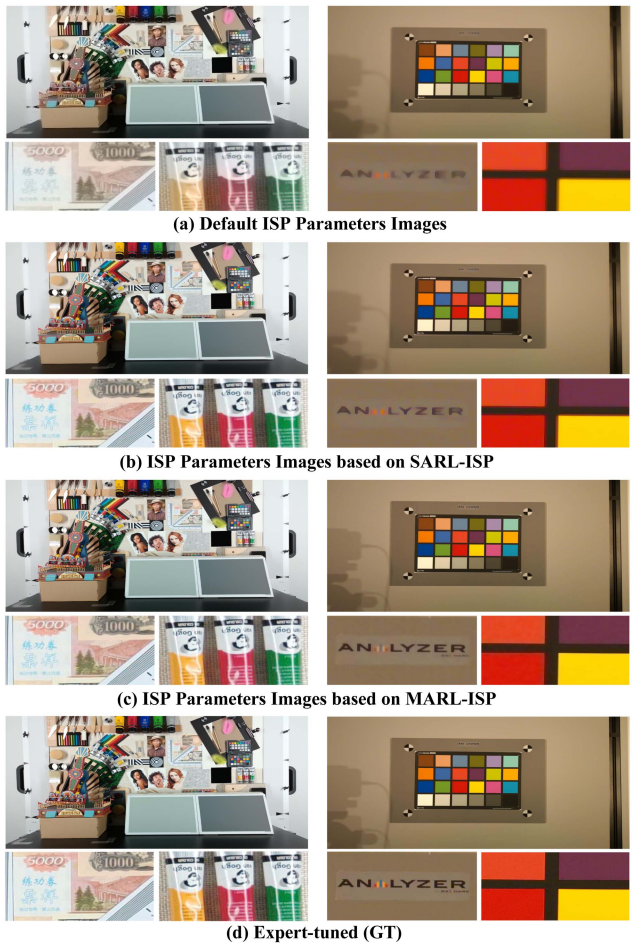

Fig. 5. Visualization examples in the image quality assessment task.

E. Ablation Study

We conducted extensive ablation studies to assess the impact of various factors on the performance of the MARL, while also adding more ablation experiments about SARL. Firstly, we verify the effectiveness of two key modules in the MARL-ISP and show visualization examples of the SPTM. Then we analyze the impact of different ISP grouping strategies. Furthermore, we perform thorough ablation experiments on the selection of reward functions and pre-trained models for downstream tasks. For the FSM, we compare the effects of multiple single features and fused features on different downstream tasks. Moreover, we analyze the optimization results of different maximum optimization step sizes and present some visual examples. Finally,

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TABLE IV
ABLATION STUDY ON THE IMPACT OF DIFFERENT COMPONENTS ACROSS THREE DOWNSTREAM TASKS, INCLUDING THE BASELINE, +DPOM, +SPTM, +FSM, AND THE COMPLETE MARL-ISP, DEMONSTRATING THE INFLUENCE OF EACH MODULE ON OVERALL PERFORMANCE

Methods	Object Detection			Instance Segmentation			Image Quality Assessment	
	mAP@0.5	mAP@0.75	mAP@0.5:0.95	mAP@0.5	mAP@0.75	mAP@0.5:0.95	SSIM	PSNR
Baseline	0.604	0.402	0.396	0.578	0.362	0.335	0.80214	30.430
+ DPOM	0.611	0.427	0.409	0.596	0.374	0.342	0.81961	31.626
+ SPTM	0.632	0.465	0.434	0.621	0.405	0.360	0.85732	33.740
+ FSM	0.623	0.460	0.428	0.617	0.403	0.358	0.84452	32.897
MARL-ISP	**0.645**	**0.484**	**0.445**	**0.646**	**0.423**	**0.379**	**0.87621**	**35.013**

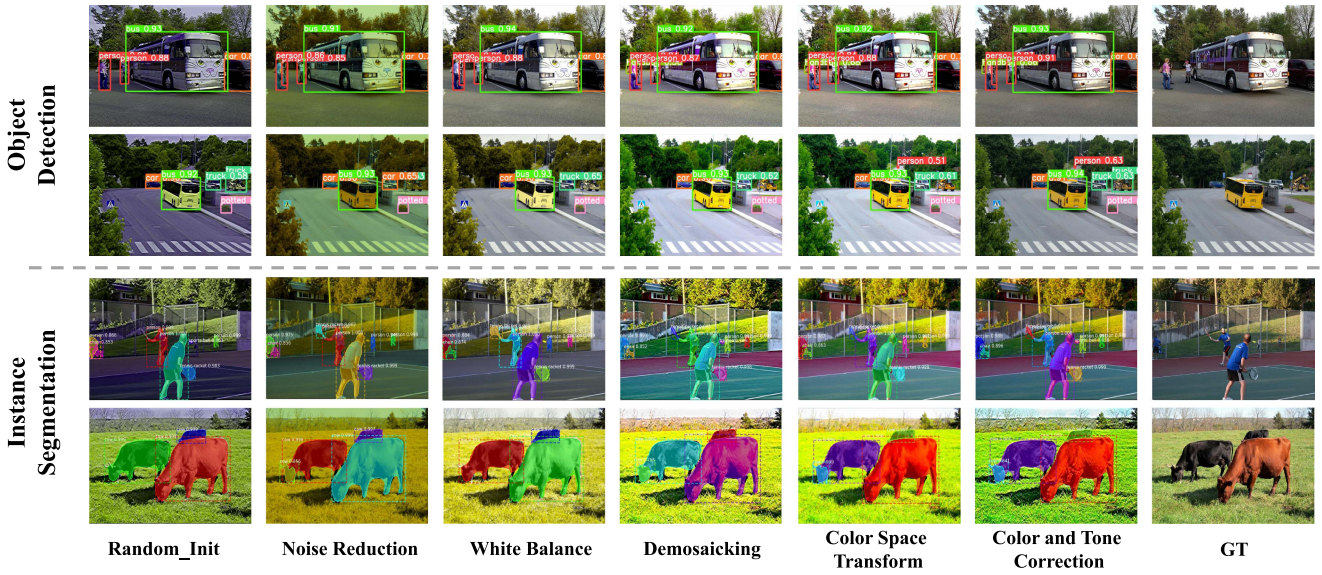

Fig. 6. Visualization examples of the sequential module optimization results on the object detection and instance segmentation. The first column represents images generated based on random parameters in the initial stage. Columns two to six are the image results generated by predicting parameters in sequence according to the module sequence structure in ISP-I. The last column shows the ground truth images.

we conduct in-depth verification and analysis of the impact of the training set size, training time, and efficiency, as well as hyperparameters in the experiments. It should be noted that the experiments related to the IQA in this section, unless otherwise specified, are conducted using ISP-II, which includes more modules and parameters.

Influence of Different Components: We use the multi-agent model without any modules as the baseline. During the experiment, we set the maximum step size to the optimal step size for different tasks, then we add the above modules respectively to verify the effect of each module. The experimental results are shown in Table IV. The integration of the DPOM achieves consistent gains over the baseline across downstream tasks, highlighting its effectiveness in MARL-ISP. The addition of the SPTM to the baseline significantly improves the model's performance in each task. This proves that the serialized parameter tuning structure effectively learns the relationship among physical ISP modules, which promotes further improvement of results. Similarly, incorporating the FSM into the baseline also improves the results across all tasks, demonstrating that it can select more suitable features for different downstream tasks. Furthermore, the MARL-ISP with all modules yields the best overall performance.

To visually demonstrate the impact of SPTM on image quality and downstream tasks, we visualize the serialized tuning results at the time step t. As shown in Fig. 6, for object detection and instance segmentation, we randomly initialize ISP parameters to accentuate visual differences over a tuning round. Then, each agent sequentially tunes the corresponding module parameters. The image corresponding to each module is generated from the raw image and all ISP parameters. The parameters of this module and its preceding modules have been predicted, while the subsequent modules remain their initial random ones. It is evident that the initialized images are of low clarity, with significant noise and severe color bias. As the parameters in each module are optimized, the images gradually become clearer and more natural, resulting in significant improvements in detection and segmentation.

The visualization results of the serialized tuning process for IQA are shown in Fig. 7, including zoomed-in regions for clearer comparison. The image generated by randomly initialized parameters is blurry and lacks details, leading to poor subjective image quality. After adjusting the parameters of each module, the image has significantly improved in terms of texture details and noise. The quality of the final RGB image is comparable to the GT image provided by experts.

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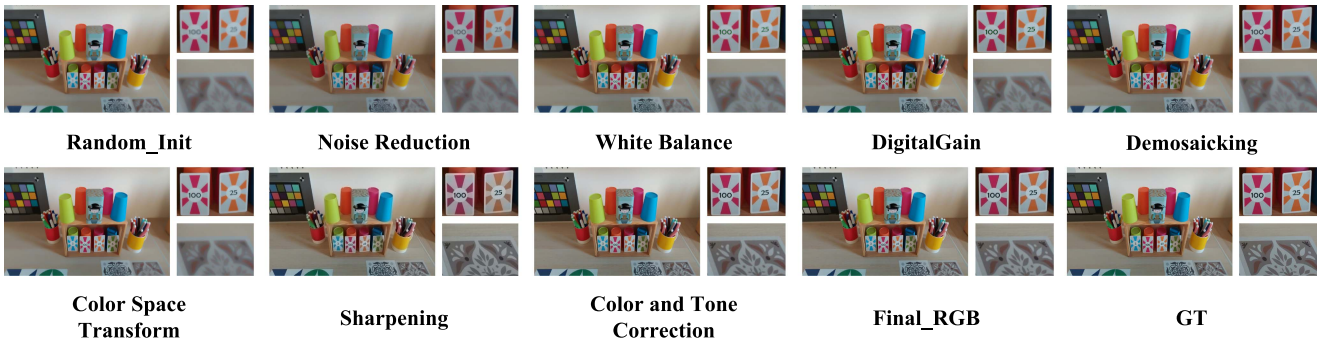

Fig. 7. Visualization examples of the sequential module optimization results based on ISP-II in the image quality assessment task.

TABLE V
ABLATION STUDY COMPARING THE EFFECTS OF DIFFERENT REWARD FUNCTIONS ON THREE DOWNSTREAM TASKS FOR SARL-ISP AND MARL-ISP

Methods	Object Detection				Instance Segmentation				Image Quality Assessment		
	Rewards	mAP@0.5	mAP@0.75	mAP@0.5:0.95	Rewards	mAP@0.5	mAP@0.75	mAP@0.5:0.95	Rewards	SSIM	PSNR
SARL-ISP	$-L_{yolo}$	**0.640**	**0.463**	**0.437**	Recalls	0.605	0.374	0.328	SSIM	**0.821**	32.384
	mAP	0.636	0.458	0.430	mAP	**0.630**	**0.405**	**0.361**	PSNR	0.808	**33.735**
MARL-ISP	$-L_{yolo}$	**0.645**	**0.484**	**0.445**	Recalls	0.623	0.406	0.346	SSIM	**0.876**	35.013
	mAP	0.638	0.480	0.441	mAP	**0.646**	**0.423**	**0.379**	PSNR	0.869	**35.963**

Influence of Grouping Strategy: The final grouping strategy chosen for the MARL-ISP is that each agent in the multi-agent model is responsible for predicting all parameters within a single ISP module. Since the ISP-II used for IQA contains relatively more modules and parameters, leading to more grouping choices, we focus on the IQA task in this section to explore the impact of different grouping strategies.

Initially, we try an average grouping strategy where each agent is responsible for predicting the same number of parameters in the ISP. The ISP-II contains a total of 114 parameters. In order to align the number of agents N_a with the number of modules, we set up 6 agents to tune 19 parameters in sequence. The parameters within the same group may belong to a single functional module of the ISP or from different functional modules. We also set N_a to 1, 3, 19, 38, and 114 for further experiments. The experimental results are shown in Table VI. When N_a is 1, all parameters are predicted by a single agent, and the number of parameters included in this module N_p is 114. It can be approximately regarded as a single-agent algorithm. However, the results are slightly worse than those of SARL-ISP shown in Table II, due to the ineffectiveness of SPTM in MARL-ISP when we set N_a to 1. We also conduct additional experiments and analysis for the MARL-ISP (single-agent) and SARL-ISP. For further details, please refer to the *Supplementary Material*. Then we increase N_a to 3, 6, 19, 38, and 114 to explore the impact of different grouping strategies on the results. It should be mentioned that we even increase N_a to 114, with each agent being responsible for predicting one parameter. The experimental results show that increasing the number of agents does not necessarily improve the model performance. It highlights the importance of considering the complex relationship among inter-group and intra-group parameters. Moreover, we also use K-Means [61] and KNN [62] to group the ISP parameters. The results show that the grouping strategies based on K-Means and KNN outperform the average grouping strategy. Ultimately, dividing parameters according to the functional modules in the ISP yields the best experimental results.

TABLE VI
ABLATION STUDY FOR DIFFERENT GROUPING STRATEGIES BASED ON ISP-II IN THE IQA TASK. THE STRATEGIES INCLUDE EVENLY DIVIDING AGENTS INTO GROUPS (AVERAGE), CLUSTERING OR NEIGHBOR-BASED GROUPING BY PARAMETER SIMILARITY (K-MEANS AND KNN), AND GROUPING ACCORDING TO ISP MODULES (MODULE)

N_a	N_p	Mode	Evalation Metric	
			SSIM	PSNR
1	[114]	Average	0.80137	31.940
3	[38, 38, 38]	Average	0.81049	31.735
6	[19, 19, ..., 19]	Average	0.84672	32.829
19	[6, 6, ..., 6]	Average	0.81693	32.101
38	[3, 3, ..., 3]	Average	0.80949	31.821
114	[1, 1, ..., 1]	Average	0.80165	31.580
6	[48, 29, 17, 14, 4, 2]	K-Means	0.83226	32.021
6	[96, 1, 6, 1, 2, 8]	KNN	0.84103	32.745
6	[41, 22, 11, 9, 25, 6]	Module	**0.87621**	**35.013**

Influence of the Reward Function: To evaluate the impact of different reward metrics, we conduct experiments on different tasks. The results are shown in Table V. For each task in SARL-ISP and MARL-ISP, we select the two representative metrics to calculate the reward. In object detection, using $-L_{yolo}$ for reward calculation resulted in better performance than using mAP. We argue that mAP as a reward may not effectively focus the model on optimizing the precision of bounding box locations,

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TABLE VII
ABLATION STUDY OF DIFFERENT PRE-TRAINED DOWNSTREAM TASK MODELS ON TWO MACHINE VISION TASKS (OBJECT DETECTION AND INSTANCE SEGMENTATION) IN MARL-ISP AND SARL-ISP

Object Detection				Instance Segmentation			
Methods	mAP@0.5	mAP@0.75	mAP@0.5:0.95	Methods	mAP@0.5	mAP@0.75	mAP@0.5:0.95
MARL-ISP							
YOLOv3 [47]	0.645	0.484	0.445	Mask R-CNN [53]	0.646	0.423	0.379
YOLOv4 [48]	**0.679**	0.496	0.451	SOLO [54]	**0.654**	**0.463**	**0.427**
YOLOv5 [49]	0.658	**0.503**	**0.466**	SAM [55]	0.453	0.292	0.264
SARL-ISP							
YOLOv3 [47]	0.640	0.463	0.437	Mask R-CNN [53]	0.630	0.405	0.361
YOLOv4 [48]	**0.661**	0.480	0.438	SOLO [54]	**0.644**	**0.413**	**0.369**
YOLOv5 [49]	0.646	**0.485**	**0.450**	SAM [55]	0.451	0.284	0.273

potentially leading to more false detections or localization errors. In contrast, using $-L_{yolo}$ as the reward enables the model to jointly fine-tune localization and classification, improving detection accuracy. In instance segmentation, using mAP as the reward achieved better results compared to recall. We believe that, in addition to identifying the targets, the model also needs to perform precise boundary segmentation. Even with a high recall, low segmentation precision still produces suboptimal segmentation outcomes. In IQA, we find that the results obtained by selecting SSIM or PSNR as the reward will perform better on the corresponding indicators, which aligns with our expectations.

Influence of Downstream Task Models: We select different pre-trained downstream models for comparisons on object detection and instance segmentation. The corresponding results obtained by our methods are reported in Table VII. For object detection, we select three mainstream methods from the YOLO series for experimentation. Both our methods and other mainstream approaches used pre-trained YOLOv3 [47] as the object detection tool. When switching to the more advanced pre-trained models, YOLOv4 [48] and YOLOv5 [49], both MARL-ISP and SARL-ISP results show significant improvements. For instance segmentation, we also compare three different mainstream methods for experiments. First, we use the pre-trained Mask R-CNN model [53] as the tool for instance segmentation. When employing the more advanced SOLO model [54], the results are superior to those of Mask R-CNN. However, testing the SAM model [55] on the MSCOCO dataset for zero-shot instance segmentation yields relatively poor performance. These experiments demonstrate the impact of different pre-trained downstream task models on the final results. Therefore, we employ the specific pre-trained models rather than other better downstream task models to ensure a fair comparison with other methods.

Influence of Different Features: The quality of features plays a significant role in the learning of the agent's strategy and the final image quality. We select and test different single and fused features for various downstream tasks to identify the most suitable input for each task. The results are shown in Table VIII. For object detection, we introduce three features: F_{yolo}, which is extracted using the YOLOv3 model, F_{resnet}, which is the feature extracted based on the ResNet-50 pre-trained network,

TABLE VIII
ABLATION STUDY ON THE IMPACT OF DIFFERENT FEATURE COMBINATIONS ON PERFORMANCE ACROSS THREE DOWNSTREAM TASKS

Task	Feature			Evalation Metric		
Detection	F_{yolo}	F_{resnet}	F_{para}	mAP@0.5	mAP@0.75	mAP@0.5:0.95
	✓	-	-	0.623	0.460	0.428
	-	✓	-	0.611	0.430	0.408
	✓	-	✓	0.623	0.464	0.432
	-	✓	✓	0.601	0.412	0.404
	✓	✓	-	**0.645**	**0.484**	**0.445**
	✓	✓	✓	0.628	0.465	0.437
Segmentation	F_{image}	F_{resnet}	F_{para}	mAP@0.5	mAP@0.75	mAP@0.5:0.95
	✓	-	-	0.617	0.403	0.358
	-	✓	-	0.598	0.382	0.347
	✓	-	✓	0.623	0.401	0.362
	-	✓	✓	0.620	0.389	0.350
	✓	✓	-	**0.646**	**0.423**	**0.379**
	✓	✓	✓	0.630	0.407	0.360
IQA	F_{resnet}	F_{para}	-	SSIM	PSNR	
	✓	-	-	**0.87621**	**35.01265**	
	✓	✓	-	0.83472	32.38271	

and F_{para}, which represents the parameter feature. We choose a single feature or a fusion of multiple features as the input of the agent network, and the results of the higher score will guide us to determine which feature is more suitable for this sub-task.

After experimental verification, F_{yolo} is proven to be the best single feature, while the fusion of F_{yolo} and F_{resnet} outperforms other feature combinations. Because the fusion of image features promotes the agent to explicitly learn ISP parameter values that enhance the clarity and texture in local image areas. For instance segmentation, we introduce F_{image}, F_{resnet}, and F_{para} respectively. F_{image} is a one-dimensional feature obtained through image conversion, expansion, and other operations. Similar to object detection, F_{image} performs best among all single features, and the fusion features of F_{image} and F_{resnet} outperforms other feature combinations. For the IQA task, we use F_{resnet} and F_{para}. The evaluation metrics focus on the difference between the image and GT. The results show that using F_{resnet} alone substantially outperforms the fusion of the two features in both SSIM and PSNR. This outcome suggests that introducing F_{para} may mislead the agents into making incorrect or suboptimal

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TABLE IX
ABLATION STUDY OF MAXIMUM OPTIMIZATION STEP T ON PERFORMANCE ACROSS THREE DOWNSTREAM TASKS, WITH STEPS OF 1, 2, 4, 8, AND 16

Max Step	Object Detection			Instance Segmentation			Image Quality Assessment	
	mAP@0.5	mAP@0.75	mAP@0.5:0.95	mAP@0.5	mAP@0.75	mAP@0.5:0.95	SSIM	PSNR
T=1	0.602	0.404	0.389	0.586	0.382	0.323	0.80541	30.56123
T=2	0.617	0.423	0.405	0.621	0.401	0.355	0.82712	32.15964
T=4	**0.645**	**0.484**	**0.445**	0.623	0.404	0.362	**0.87621**	**35.01265**
T=8	0.634	0.465	0.433	**0.646**	**0.423**	**0.379**	0.84974	33.53784
T=16	0.621	0.442	0.417	0.636	0.397	0.374	0.83154	32.12651

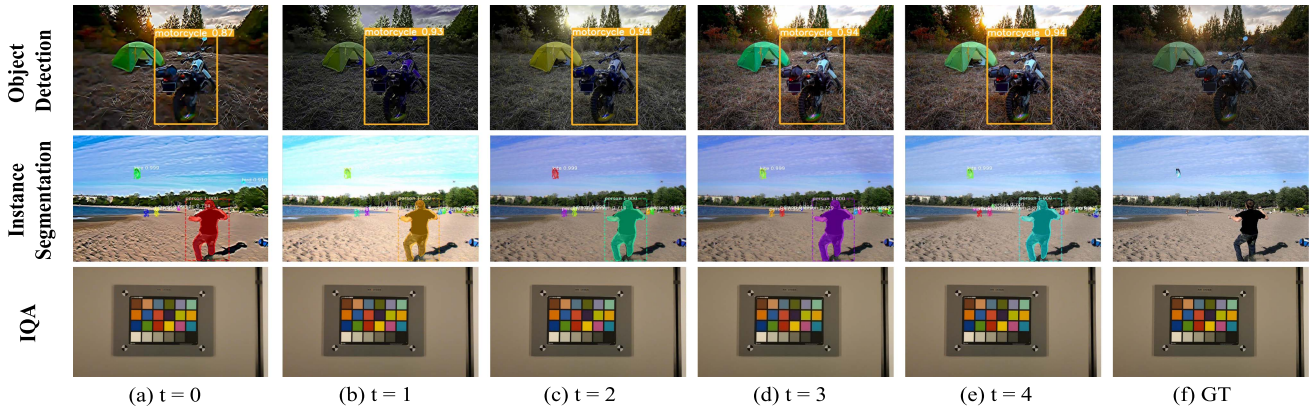

Fig. 8. Visualization examples of the sequential optimized results during the validation phase on three downstream tasks, i.e., object detection (top lines), instance segmentation (middle lines), and IQA (bottom lines), respectively. (a) represents images generated based on random parameters in the initial stage. (b)-(e) indicate the images produced by our models after ISP parameter optimization at each step. (f) shows the ground truth images.

decisions, negatively impacting the accuracy of ISP parameters predictions and overall image quality.

Influence of the Maximum Optimization Step T: As illustrated in Table IX, we explore the effect of the maximum optimization step T. As T gradually increases, the model's performance improves, demonstrating that multi-step parameter optimization effectively helps the agents better predict ISP parameters. For object detection and IQA, our model achieves the best results when T is set to 4. For instance segmentation, which requires higher image quality in detailed regions, the model achieves the best results when the max step is set to 8. After experimental analysis, we find that the complexity of the task is the main factor influencing the differences in max step across different tasks. Additionally, we also find that the choice of reward function also affects the max step. We have added more experiments to thoroughly discuss the above issues in the *Supplementary Material*.

As shown in Fig. 8, we present visual examples of different tasks with a maximum step size T of 4. The results demonstrate that increasing the number of steps improves the detection performance and segmentation accuracy. The same effect is seen in IQA. It enhances texture clarity and reduces noise, demonstrating the effectiveness of our tuning process.

Influence of the Training Set Data Size: Unlike traditional machine learning methods, we do not require a large amount of training data to ensure model effectiveness. We conduct ablation experiments to explore the impact of training data size on model performance. The results are shown in Table X. We reduce the training data from the original 1000 images to 800, 500, 300, and 100, respectively. It is clear that the performance of the MARL with 800 training images is nearly

TABLE X
ABLATION STUDY OF VARYING THE NUMBER OF TRAINING IMAGES ON PERFORMANCE ACROSS THREE DOWNSTREAM TASKS

Task	Training Images	Evalation Metric		
Detection	-	mAP@0.5	mAP@0.75	mAP@0.5:0.95
	100	0.526	0.365	0.325
	300	0.572	0.395	0.369
	500	0.603	0.426	0.397
	800	**0.645**	**0.484**	**0.445**
	1000	0.644	0.482	0.445
Segmentation	-	mAP@0.5	mAP@0.75	mAP@0.5:0.95
	100	0.503	0.326	0.305
	300	0.538	0.361	0.327
	500	0.589	0.392	0.351
	800	**0.646**	**0.423**	**0.379**
	1000	0.645	0.423	0.378
IQA	-	SSIM	PSNR	
	50	0.84783	32.07459	
	100	0.86215	33.25163	
	175	**0.87621**	**35.01265**	

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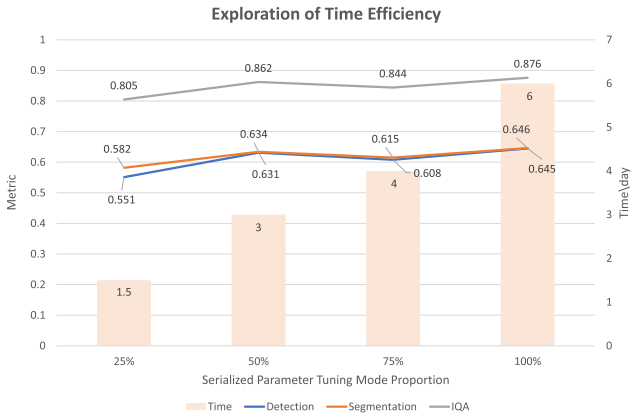

Fig. 9. Comparison of the time and effect of different serialization parameter tuning mode proportions on various downstream tasks.

comparable to that with 1000 training images. However, performance degrades significantly with further data reduction. For the IQA task, the available training data are inherently limited. Experimental results demonstrate that the best results can be achieved as the number of training scenes increases to 175.

Analysis of Model Training Time and Efficiency: The serialized structure of the SPTM necessitates that each agent tunes its parameters based on the parameters and image information provided by the preceding agent. The entire pipeline is required to repeat the hardware ISP and feature extraction operations for n times (*i.e.*, the number of modules) within one step, which incurs enormous time consumption.

To address the problem, we propose a parallel prediction mode to balance the effect and efficiency of the model. In this mode, default parameters and the raw images are fed into the ISP and the features are extracted only once from the initial image. Then these features are used as input for all agents. It significantly reduces the exploration time. We combine the two prediction modes in different proportions during the model exploration phase. Our objective is to reduce training time while preserving the effectiveness of the model. As shown in Fig. 9, the abscissa is the proportion of SPTM in the exploration phase, which is set to 25%, 50%, 75%, and 100%, respectively. The left and right ordinates are the evaluation metrics (*i.e.*, mAP and SSIM) and the training time of the model, respectively. The curves represent the performance of the corresponding tasks across different proportions of the serialized prediction mode. The results show that the convergence time is positively correlated with the proportion of the serialized mode. When the proportion is 50%, the performance is comparable to that at 100% across multiple metrics. It indicates that our exploration mode can reduce training time consumption to about half of the previous amount while ensuring effectiveness.

Influence of Hyperparameters: In this section, we discuss two main hyperparameters. The hyperparameter ω controls the proportion of step reward in the total reward of the current agent. Therefore, we conduct ablation experiments on the hyperparameter ω. The results are shown in Table XI. When ω is 0.2, the model performs the best in three tasks. We believe the reason is that a larger proportion of step reward diminishes the agent's responsiveness to the current module reward.

TABLE XI
ABLATION STUDY ON THE IMPACT OF HYPERPARAMETERS ω AND ϵ ON THREE DOWNSTREAM TASKS

Task	Value(ω)	Metric	Value(ϵ)	Metric
Detection	ω	mAP@0.5	ϵ	mAP@0.5
	0.1	0.640	0.1	0.643
	0.2	**0.645**	0.2	**0.645**
	0.5	0.624	0.5	0.631
	1.0	0.627	–	–
Segmentation	ω	mAP@0.5	ϵ	mAP@0.5
	0.1	0.637	0.1	0.637
	0.2	**0.646**	0.2	**0.646**
	0.5	0.612	0.5	0.618
	1.0	0.622	–	–
IQA	ω	SSIM	ϵ	SSIM
	0.1	0.865	0.1	0.862
	0.2	**0.876**	0.2	**0.876**
	0.5	0.853	0.5	0.847
	1.0	0.845	–	–

The hyperparameter ϵ serves as a clip operation in the actor network's maximization objective formula. It regulates the magnitude of policy updates. The larger or smaller ϵ will cause the policy update to be too large or too small, thus affecting the convergence time and model performance. Finally, the experimental effect is optimal when ϵ is set to 0.2.

V. LIMITATIONS AND FUTURE WORK

Compared to DL-based parameter optimization methods, our model requires fewer training samples to achieve better results. However, our method involves ISP configurations that differ from the ISP proxy in DL-based methods. The limitation of this mode is that it inevitably increases the execution time of generating ISP images. Nonetheless, we have optimized this aspect at the algorithmic level to reduce the time cost. Moreover, since IQA is a very challenging problem, we acknowledge the limitation of using traditional IQA metrics (*i.e.*, SSIM and PSNR) to score and guide the agent in our paper. As the field of IQA advances, we can integrate them into our method to train an ISP parameter optimization agent that aligns with IQA models or human preferences. Multi-dimensional IQA models can guide the agents in balancing parameter optimization across different modules and making trade-offs among multiple objectives, which further enhances the analogy to the human expert workflow. Future work will also focus on enhancing tuning efficiency for real-time applications and adapting the video ISP parameters to handle complex scenarios. More discussion and prospective studies are provided in the *Supplementary Material*.

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VI. Conclusion

This paper pioneers the use of reinforcement learning to address the hardware ISP parameter tuning problem, and proposes a single-agent-based ISP parameter optimization model. Recognizing the sequential structure of hardware ISP and the coupling relationships among their parameters, we further propose a sequential ISP framework based on collaborative multi-agent RL. Abundant experimental results show that our serialized multi-agent approach improves model performance and enhances the interpretability of the overall framework. Moreover, our models achieve excellent performance with less training data than other mainstream methods.

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